



King Abdulaziz University

(Solution)

Faculty of Computing & Information Technology

Computer Science Department

Mid Exam (Sunday March 24, 2024)

CPCS-223 (Analysis and Design of Algorithms)

Spring 2024

Total Marks: 100

Time: 80 minutes

Student ID: _____

Student Name: _____

Instructors:

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Question	SO	Max. Marks	Obt. Marks
01	1	5	
02	1	5	
03	1	5	
04	2	5	
05	1	5	
06	1	5	
Total Marks:		30	

Question 1: MCQs (5 Marks, SO: 1)

(0.5 point each)

Choose the correct option.

- (i) Which of the following complexity classes represents the minimum running time?
- (a) $O(n)$
 - (b) $O(n \log n)$
 - (c) **$O(\log n)$**
 - (d) $O(\lfloor n \rfloor)$
- (ii) Which of the following is not a part of loop invariant?
- (a) Initialization
 - (b) **Test Condition**
 - (c) Maintenance
 - (d) Termination
- (iii) Which of the following assertions is **False**?
- (a) $n(n-1) \in \wedge(n^2)$
 - (b) $n(n+1)/2 \in O(n^3)$
 - (c) $n(n-1)/2 \in o(n^3)$
 - (d) **$n(n+1)/2 \in \omega(n^3)$**
- (iv) Which of the following statements is **incorrect**?
- (a) $\log n \in O(n \log n)$
 - (b) $\lfloor n \rfloor \in \setminus(\lfloor n \rfloor)$
 - (c) $n^3 \in \wedge(n)$
 - (d) **$3^n \in O(2^n)$**
- (v) Function 'g' is an upper bound for function 'f', if and only if, for all 'x'
- (a) $g(x) \delta f(x)$
 - (b) **$f(x) \delta g(x)$**
 - (c) $f(x) \in \wedge(g(x))$
 - (d) $g(x) \in O(f(x))$
- (vi) Which of the following recurrence relations is **incorrect**, for various versions of Quick Sort algorithms?
- (a) $T(n) = T(0) + T(n-1) + \setminus(n)$
 - (b) **$T(n) = 2 T(n/2) + \setminus(1)$**
 - (c) $T(n) \delta 2 T(n/2) + \setminus(n)$
 - (d) $T(n) \delta T(9n/10) + T(n/10) + \setminus(n)$

- (vii) For which of the following conditions, Master theorem **cannot** be used?
- (a) $T(n)$ is not a monotonic function
 - (b) $f(n)$ is not a polynomial
 - (c) $a < 1$
 - (d) a is a constant**
- (viii) For functions 'f' and 'g', the condition $0 \leq f(n) < c g(n)$ for all $n \geq n_0$, represents the asymptotic notation
- (a) Little-Omega (i.e. ω)
 - (b) Little-Oh (i.e. o)**
 - (c) Capital Omega (i.e. $\wedge$)
 - (d) Big-Oh (i.e. O)
- (ix) For travelling salesman algorithm with 'n' vertices, using exhaustive search technique, the number of circuits with only distinct values is
- (a) $n!$
 - (b) $(n - 1)!$
 - (c) $1/2 * (n + 1)!$
 - (d) $1/2 * (n - 1)!$**
- (x) Which of the following is the correct recurrence relation for the Tower of Hanoi algorithm?
- (a) $T(n) = 2 T(n - 1) + 1 ; T(1) = 1$**
 - (b) $T(n) = 2 T(n - 1) + 1 ; T(1) = 0$
 - (c) $T(n) = 2 T(n - 1) + n ; T(1) = 1$
 - (d) $T(n) = 2 T(n - 1) + n ; T(1) = 0$

Question 2 (5 Points, SO: 1)

For which values of n_0 and c , the following condition holds?

$$(5n^2 - 19n) \in \Omega(n^2)$$

Solution:

Let $f(n) = (5n^2 - 19n)$ and $g(n) = n^2$. We are required to find a positive constant c and non-negative integer n_0 such that:

$$f(n) \geq cg(n) \text{ for every } n \geq n_0$$

So, consider $(5n^2 - 19n) \geq cn^2$

Divide both sides by n^2

We obtain:

$$5 - 19/n \geq c$$

This is possible only if we take $n_0 = 19$. Then we shall get $c = 4$

Hence, it is proved for $c=4$ and $n_0=19$

error = -1 pt

Question 3 (5 Points, SO: 1)

Consider the following recursive algorithm:

ALGORITHM : Sample($A[0..n - 1]$)

//Input: An array $A[0..n - 1]$ of real numbers

```
if  $n = 1$ 
    return  $A[0]$ 
else
    temp  $\leftarrow$  Sample( $A[0..n - 2]$ )
    if temp  $\in A[n - 1]$ 
        return temp
    else return  $A[n - 1]$ 
```

- a) What does this algorithm compute?
- b) what is the basic operation of this algorithm?
- c) Set up a recurrence relation for the algorithm's basic operation and then solve it, using substitution (forward or backward) method.

Solution:

- a) It returns largest element of the input array 'A' (1 point)
- b) The basic operation is comparison (1 point)
- c) The recurrence relation is $T(n) = T(n - 1) + 1$; $T(1) = 1$ (3 points)
 - For $n = 1$, $T(1) = 1$ (Base Case) 1pts
 - For $n = 2$, $T(2) = T(2-1)+1$ 1pts

$$T(2) = T(1) + 1 = 1 + 1 = 2$$

- For $n = 3$, $T(3) = T(3-1) + 1$

$$T(3) = T(2) + 1 = 2 + 1 = 3$$

.....

- For $n = k$, $T(k) = T(k-1) + 1 = (k - 1) + 1 = k$

So, from symmetry we can conclude that running time is $T(n) = n$

1pts

Question 4 (5 Points, SO: 2)

Design a recursive algorithm for computing the sum of cubes of 'n' positive integers i.e.

$$\text{SumCubes}(n) = 1^3 + 2^3 + 3^3 + \dots + n^3$$

Solution:

ALGORITHM : Computing sum of cubes of 'n' positive integers

//Input: A positive integer 'n'

//Output : The sum of first 'n' cubes

```
SumCubes(n) {
    if n == 1
        return 1 ;    2 pts
    else
        return SumCubes(n - 1) + n*n*n ;    3 pts
}
```

Question 5 (5 Points, SO: 1)

Using Master theorem, solve the following recurrence relations

a) $T(n) = 16 T(n/2) + n^2$ (2 points)

b) $T(n) = T(2n/3) + 1$ (3 points)

Solution:

a) Here $a = 16$, $b = 2$, $f(n) = n^2$

$$\begin{aligned} \text{We first find } \log_b a &= \log_2 16 = \log_2 2^4 \\ &= 4 \log_2 2 = 4 \end{aligned}$$

$$\text{Hence } n^{\log_b a} = n^4$$

So, we see that

$$f(n) \leq O(n^{\log_b a}) \text{ because } n^2 \leq n^4$$

Therefore case 1 applies

$$\text{So, } T(n) = \Theta(n^{\log_b a})$$

$$\text{Or } T(n) = \Theta(n^4)$$

b) Here $a = 1$, $b = 3/2$, and $f(n) = 1$

$$\text{We shall first find } \log_b a = \log_{3/2} 1 = 0$$

$$\text{So, } n^{\log_b a} = n^0 = 1$$

$$\text{From here we see that } f(n) = \Theta(n^{\log_b a})$$

$$\text{because } 1 = 1$$

Therefore case 2 applies

$$\text{So, } T(n) = \Theta(n^{\log_b a} \lg n)$$

$$\text{Or } T(n) = \Theta(1 \cdot \lg n)$$

Or $T(n) = \Theta(\lg n)$

Question 6 (5 Points, SO: 1)

Apply the following algorithm of Bubble Sort on the given data set and show the contents of the array after every iteration of outer loop. Furthermore, find the running time of the algorithm.

ALGORITHM *BubbleSort*($A[0..n - 1]$)
//Sorts a given array by bubble sort
//Input: An array $A[0..n - 1]$ of orderable elements
//Output: Array $A[0..n - 1]$ sorted in nondecreasing order
for $i \leftarrow 0$ to $n - 2$ do
 for $j \leftarrow 0$ to $n - 2 - i$ do
 if $A[j + 1] < A[j]$ swap $A[j]$ and $A[j + 1]$

Solution :

	0	1	2	3	4	5	6	7	8
Initial	38	27	43	3	9	82	10	22	30
1 st	27	38	3	9	43	10	22	30	82
2 nd	27	3	9	38	10	22	30	43	82
3 rd	3	9	27	10	22	30	38	43	82
4 th	3	9	10	22	27	30	38	43	82
5 th	3	9	10	22	27	30	38	43	82
6 th	3	9	10	22	27	30	38	43	82
7 th	3	9	10	22	27	30	38	43	82
8 th	3	9	10	22	27	30	38	43	82
Final Values	3	9	10	22	27	30	38	43	82